

A NEURON ENRICHED MEG3 LNCRNA ISOFORM ORGANIZES ELAVL ENRICHED NUCLEAR ASSEMBLIES TO SUPPORT LONG NEURONAL TRANSCRIPT

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Abstract

Long neuronal genes contain numerous exons and exceptionally long introns, making their transcripts particularly challenging to process. However, the mechanisms that ensure the proper maturation of these genes remain poorly understood. Through transcriptomic analyses, we identified a long noncoding RNA isoform, Meg3^{MN1}, that is maintained during neuronal aging and may play an important role in regulating long neuronal gene expression. Analysis of chromatin isolation by RNA purification (ChIRP-seq) and RNA pulldown data revealed that Meg3^{MN1} associates with members of the ELAVL RNA-binding protein family on chromatin and forms Meg3^{MN1}-ELAVLs condensates in spinal motor neurons. Reduction of Meg3^{MN1} disrupted the formation of Meg3^{MN1}-ELAVLs condensates and the chromatin association of ELAVL proteins, leading to aberrant RNA-processing events. These defects may consequently compromise the expression of long neuronal genes and contribute to neuronal dysfunction. Together, our findings support a model in which Meg3^{MN1} promotes the nuclear localization and chromatin recruitment of ELAVL proteins, thereby facilitating the mRNA processing events of long neuronal genes. This study reveals a potential mechanism through which a long noncoding RNA coordinates RNA-binding proteins to maintain neuronal gene expression.

Introduction

Long neuronal genes are difficult to process.

Many exceed 100 kb and contain numerous exons and long introns.

RBP's help regulate long neuronal transcripts.

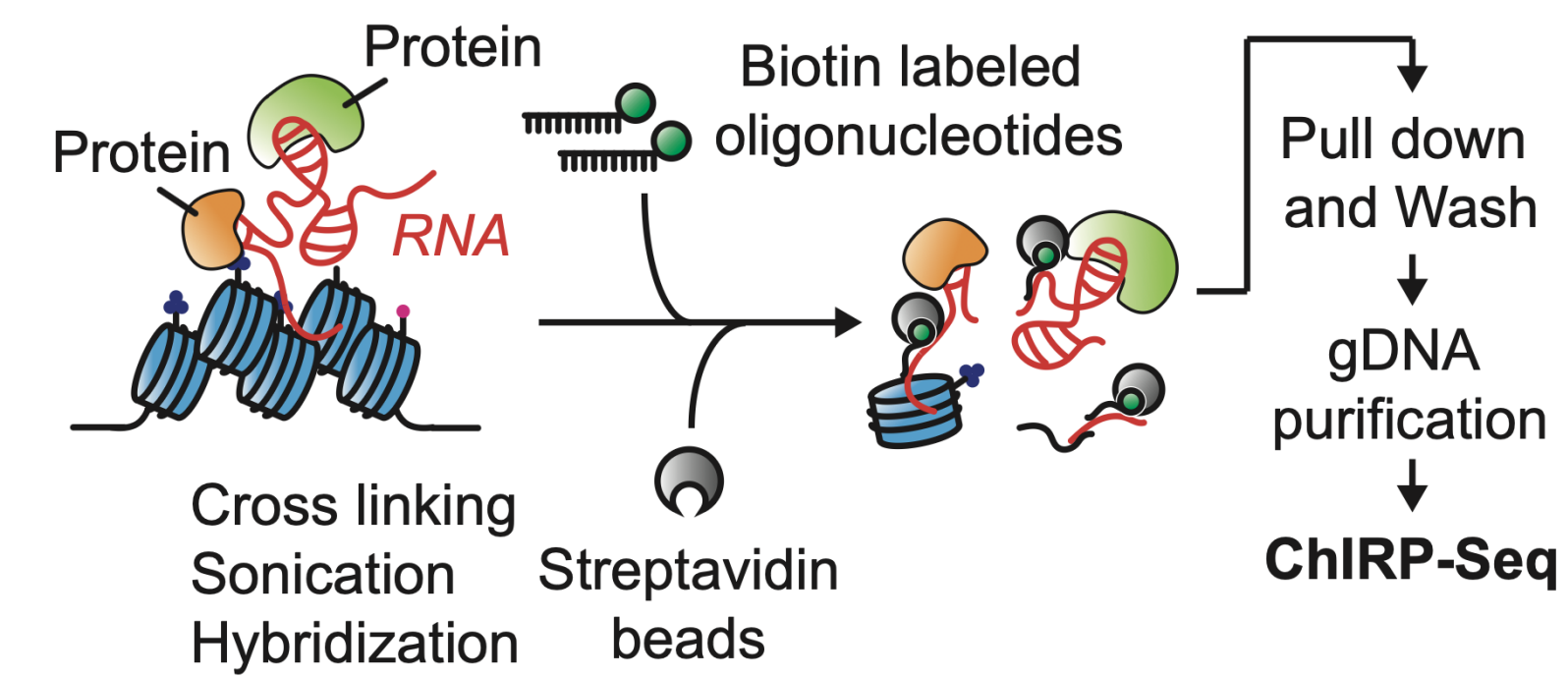
ELAVL proteins and SFPQ contribute to RNA stability, splicing, and transcription.

LncRNAs can organize RNA-processing machinery.

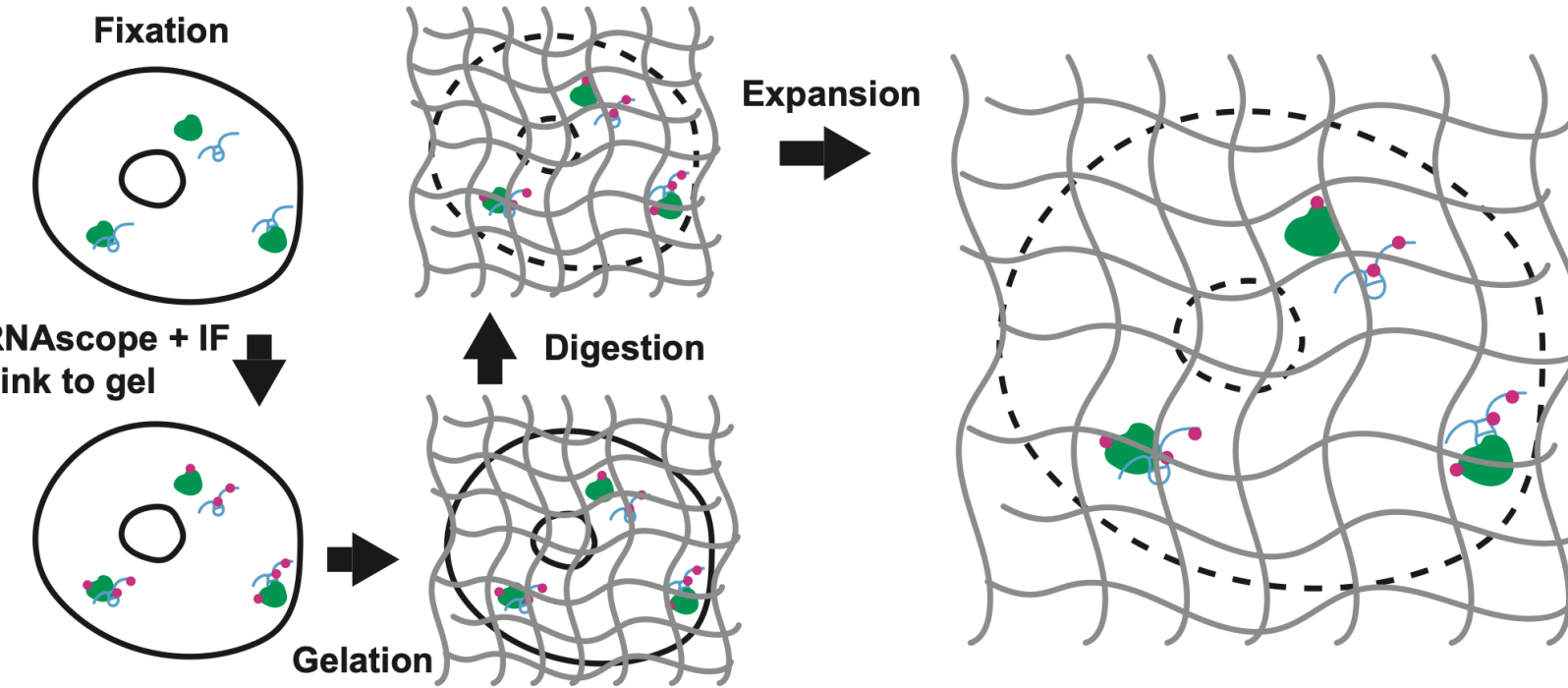
They associate with chromatin and scaffold RNA-protein complexes.

Methods

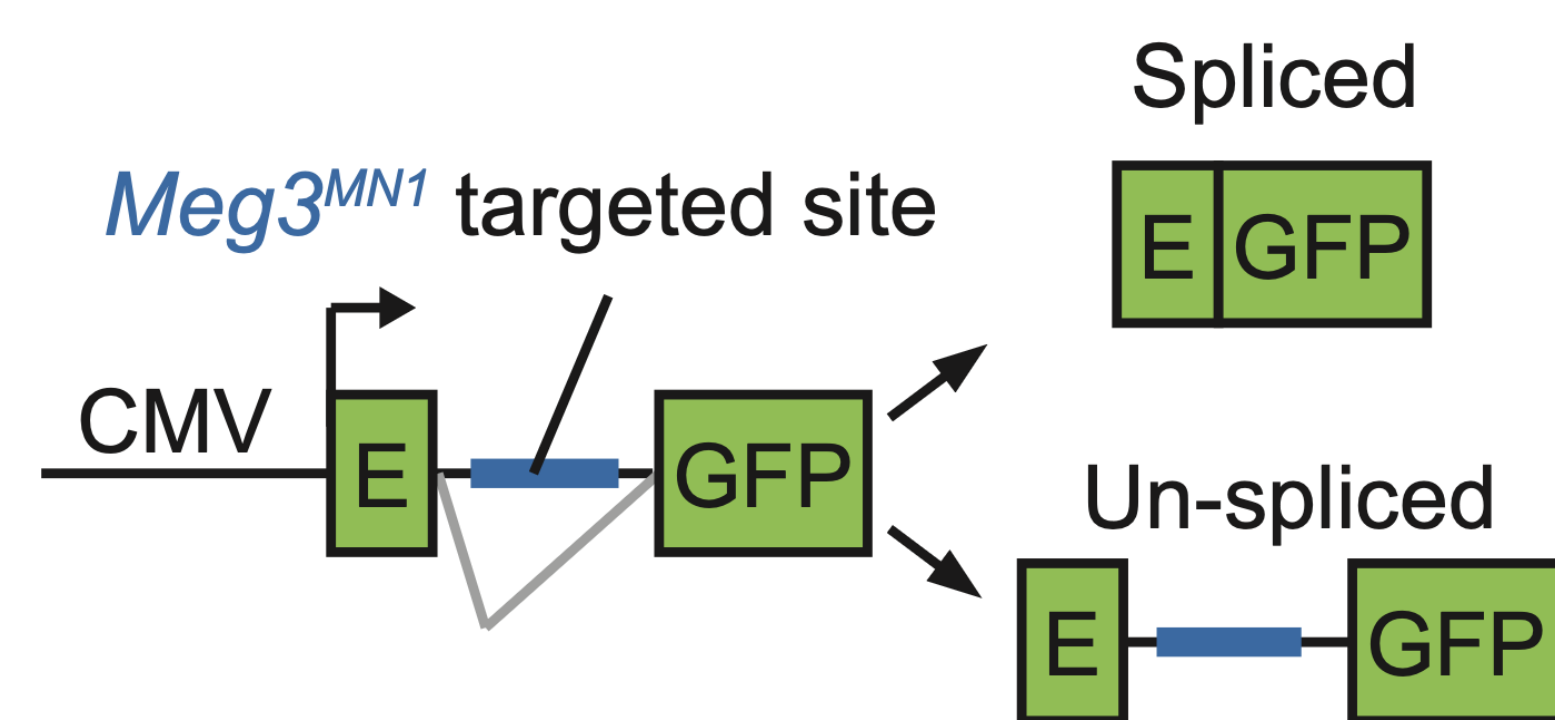
ChIRP-seq + RNA pulldown-MS / RIP



↓ RNAscope + Expansion Microscopy



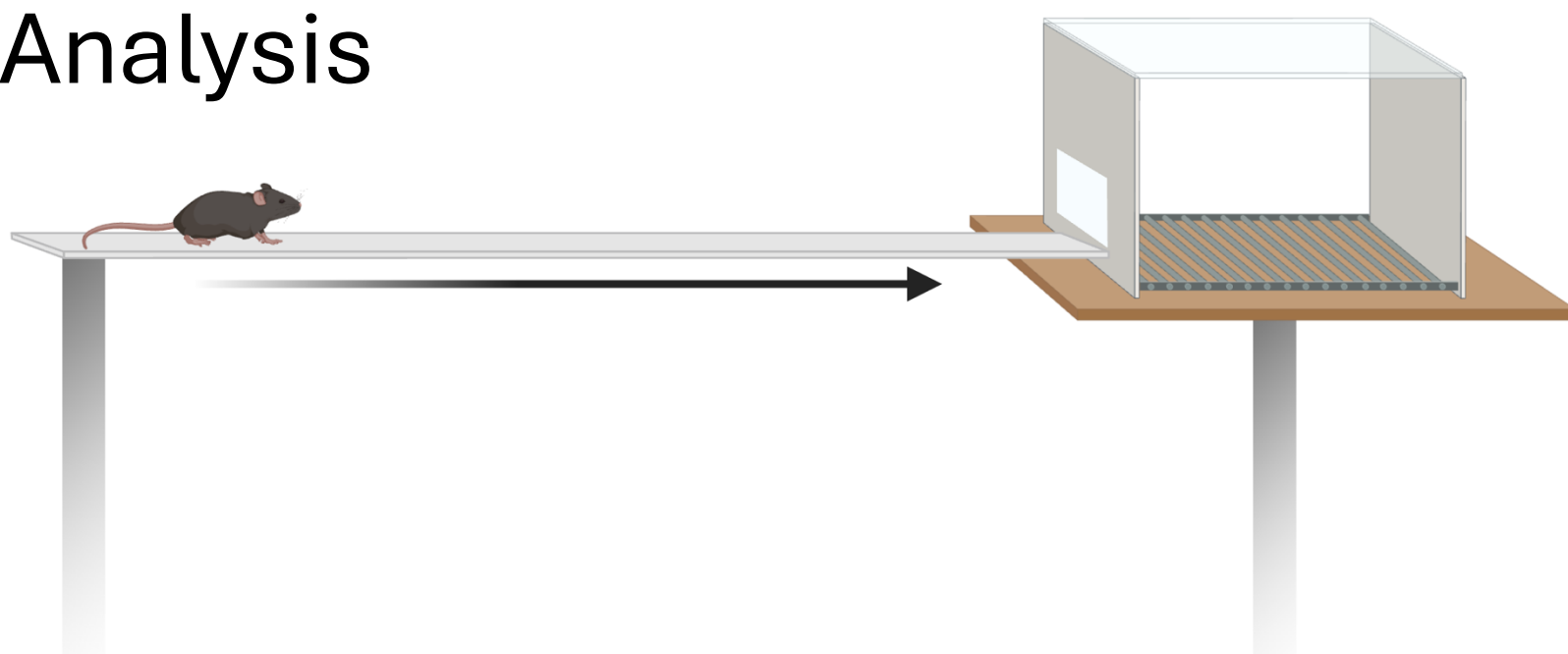
↓ Mini-GFP Splicing Reporter



↓ Meg3^MN1 KO + RNA-seq

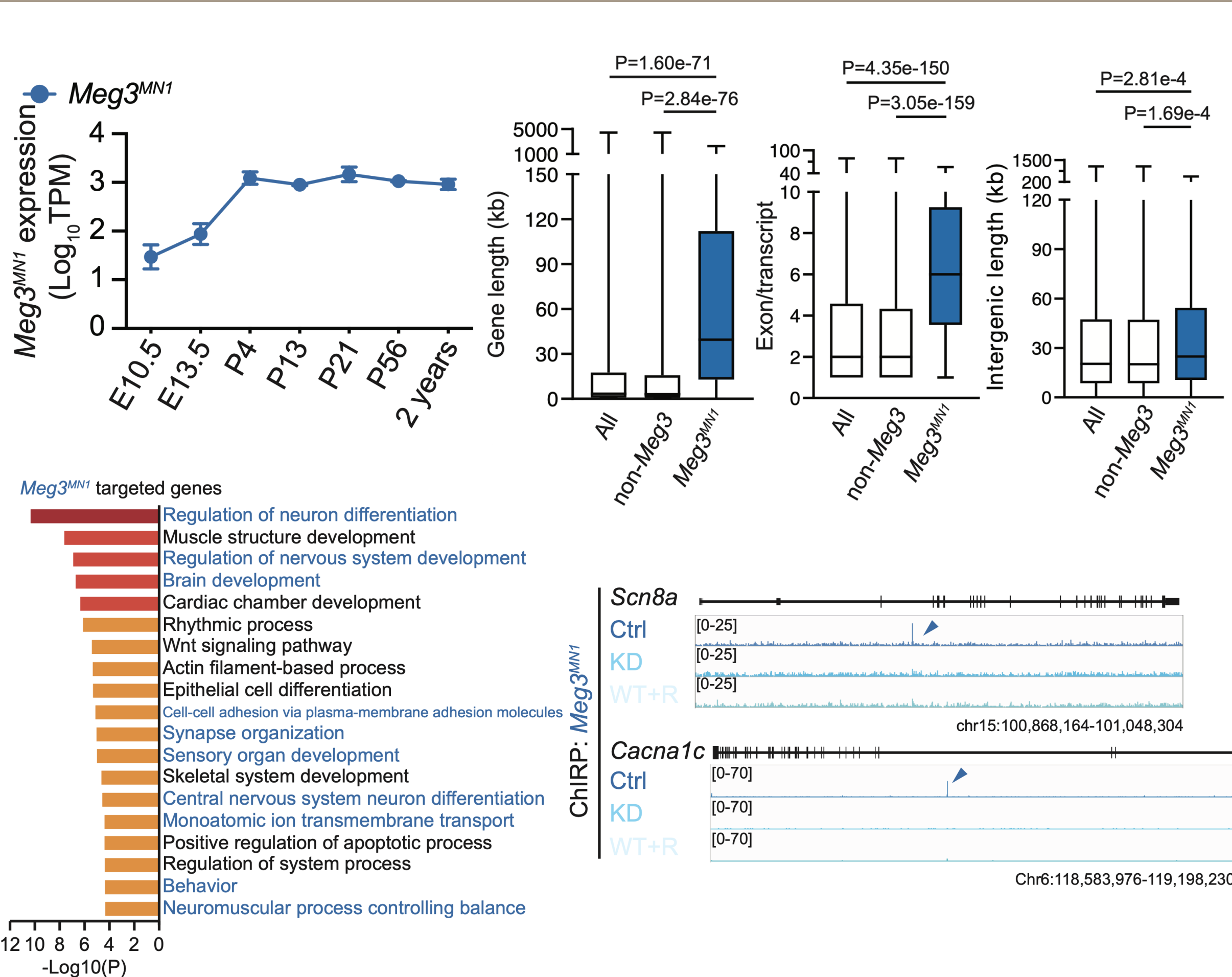


↓ Mouse Behavioral & Motor Neuron Analysis

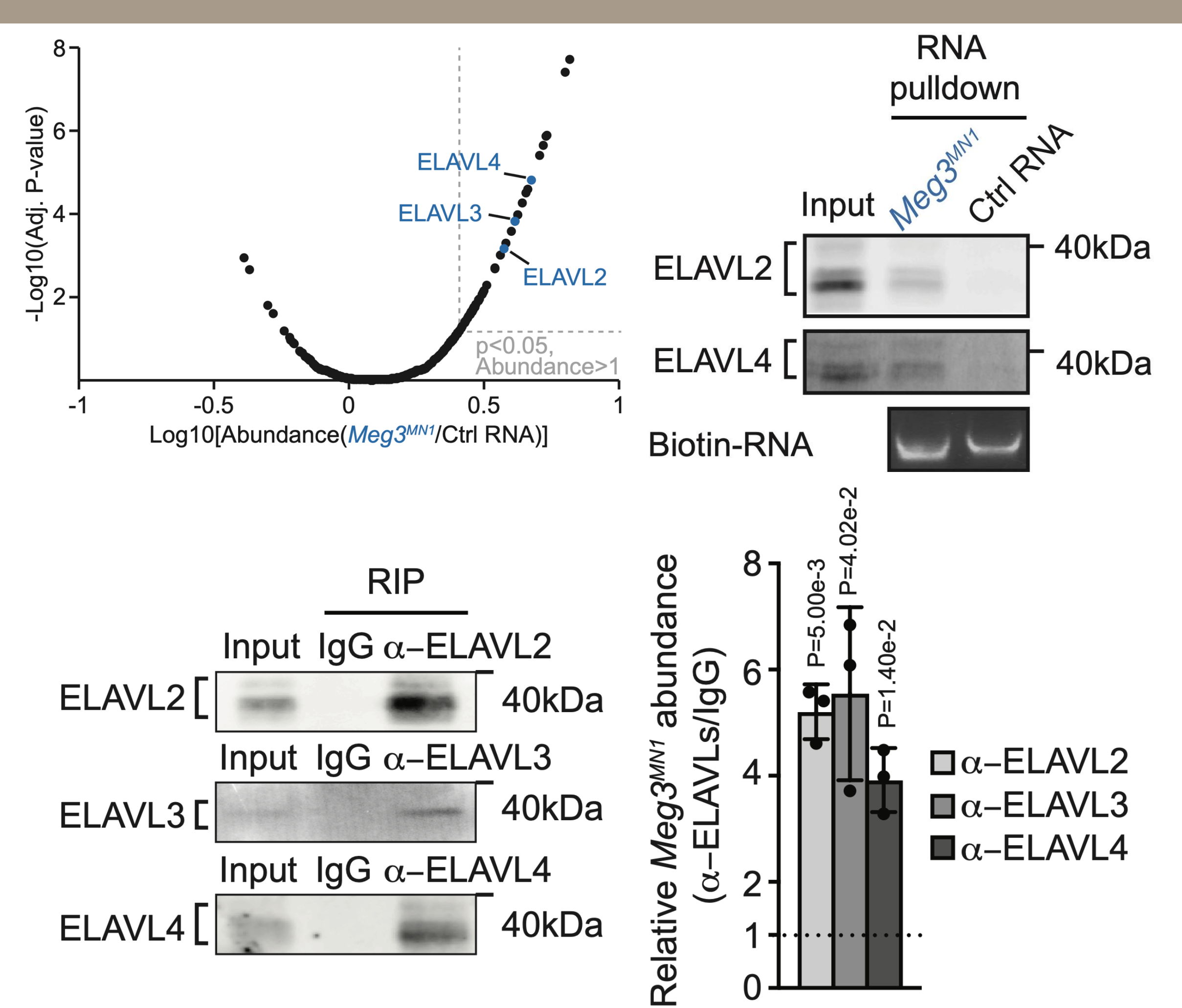


Results

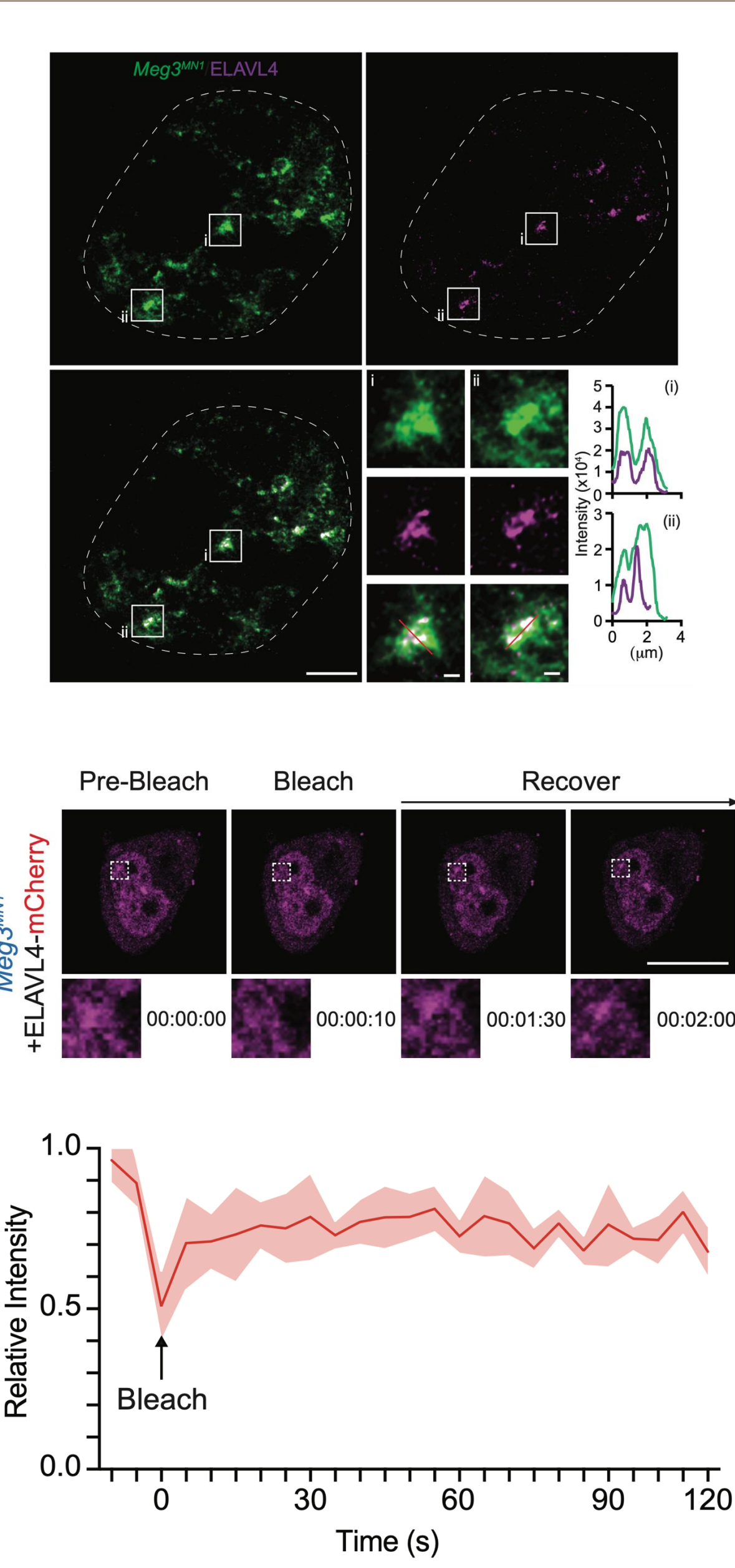
Meg3^{MN1} is maintained in neurons and targets long neuronal genes



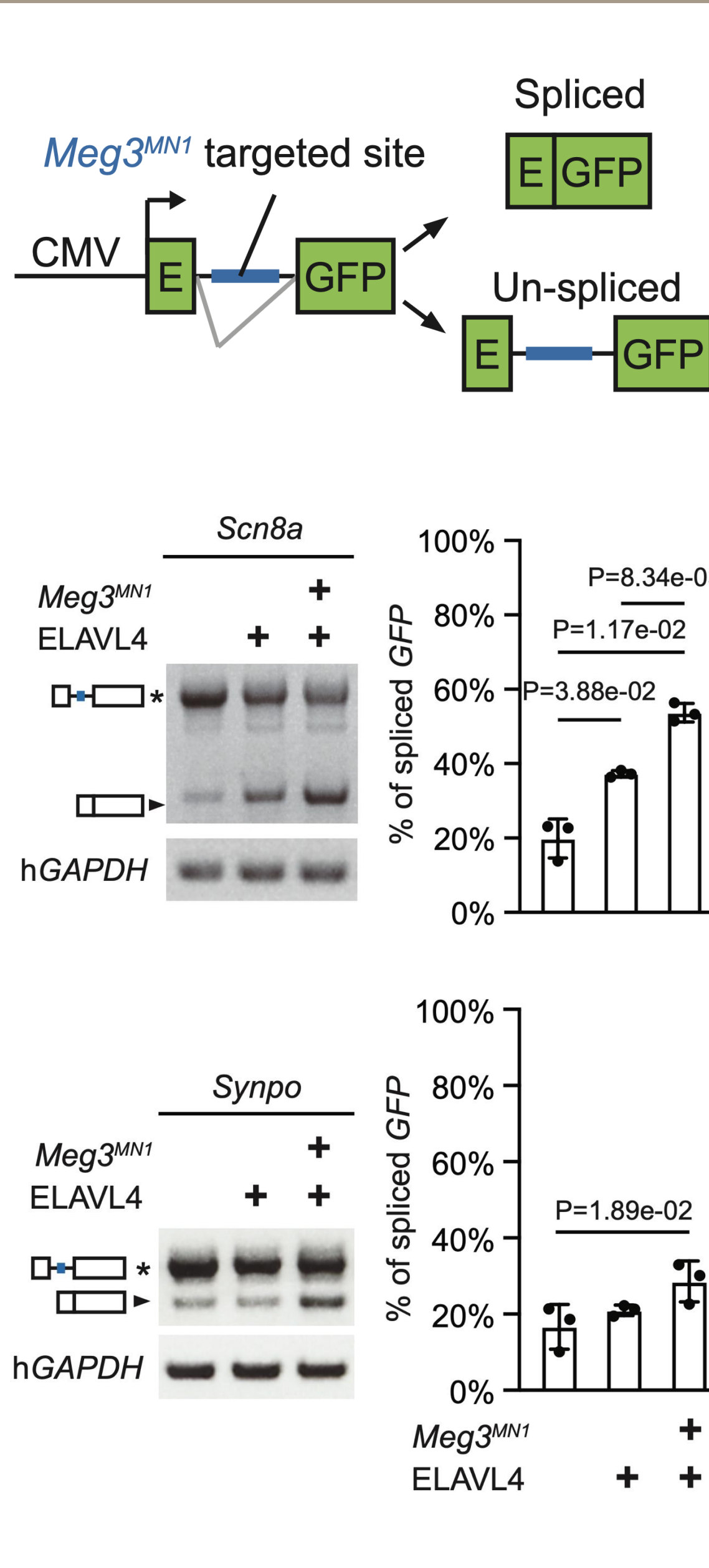
Meg3^{MN1} forms an RNP complex with neuronal ELAVL proteins



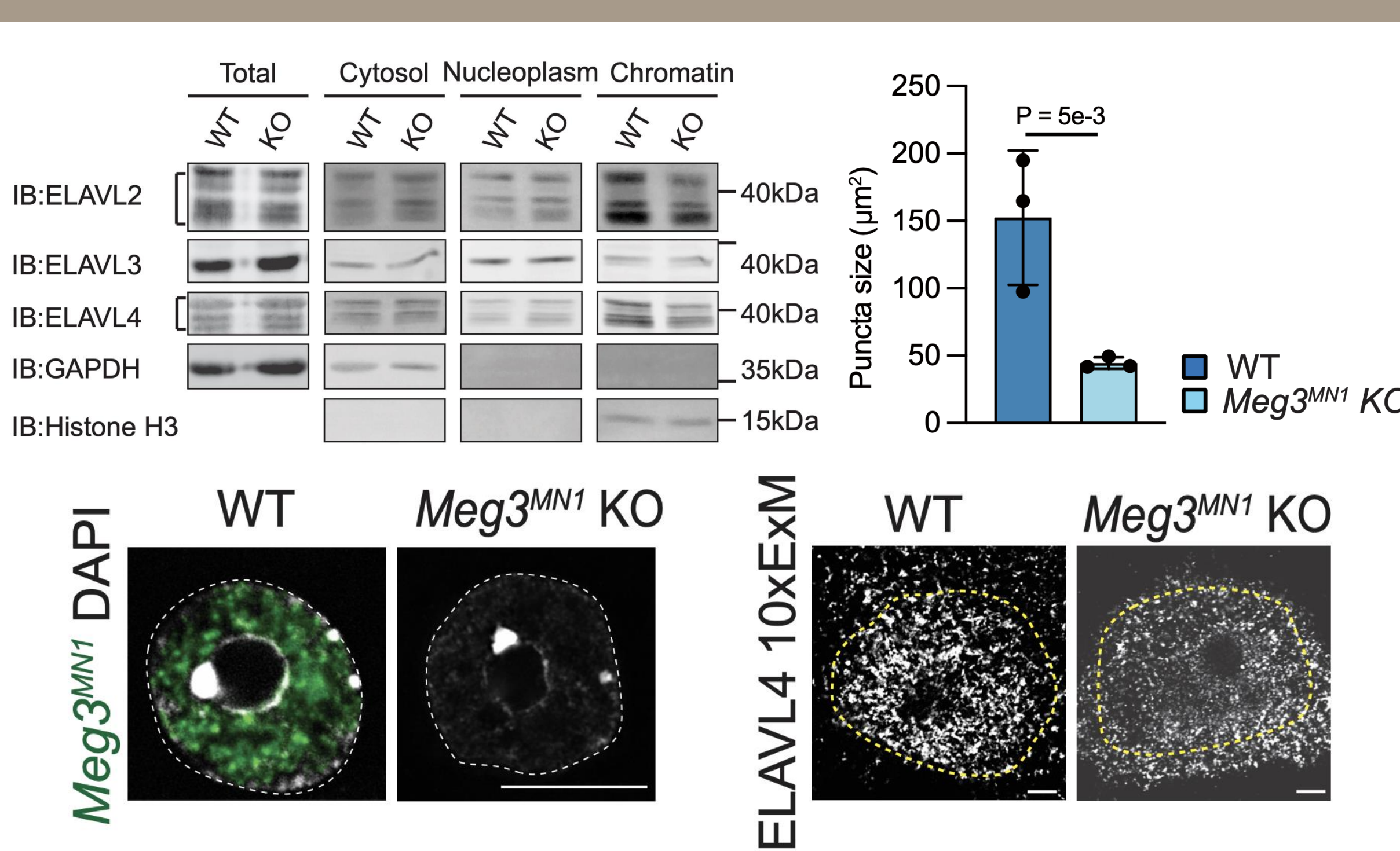
Meg3^{MN1} & ELAVL colocalize in nuclear assemblies



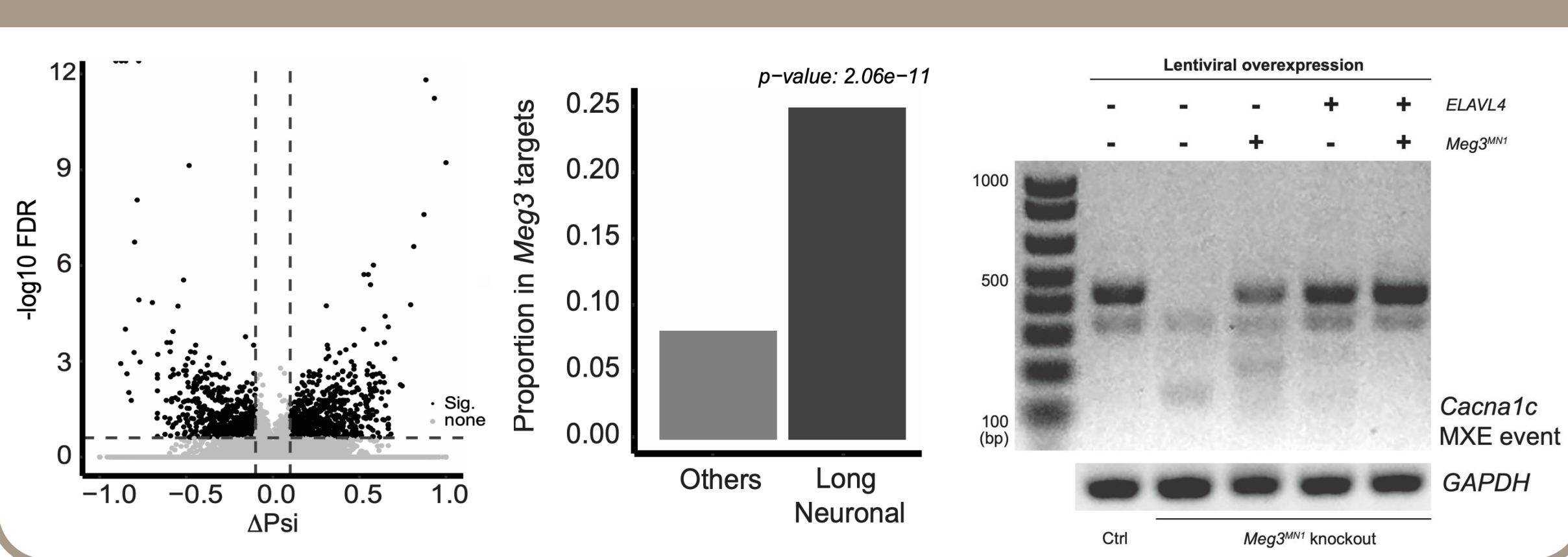
Meg3^{MN1}-ELAVL4 promotes target RNA splicing



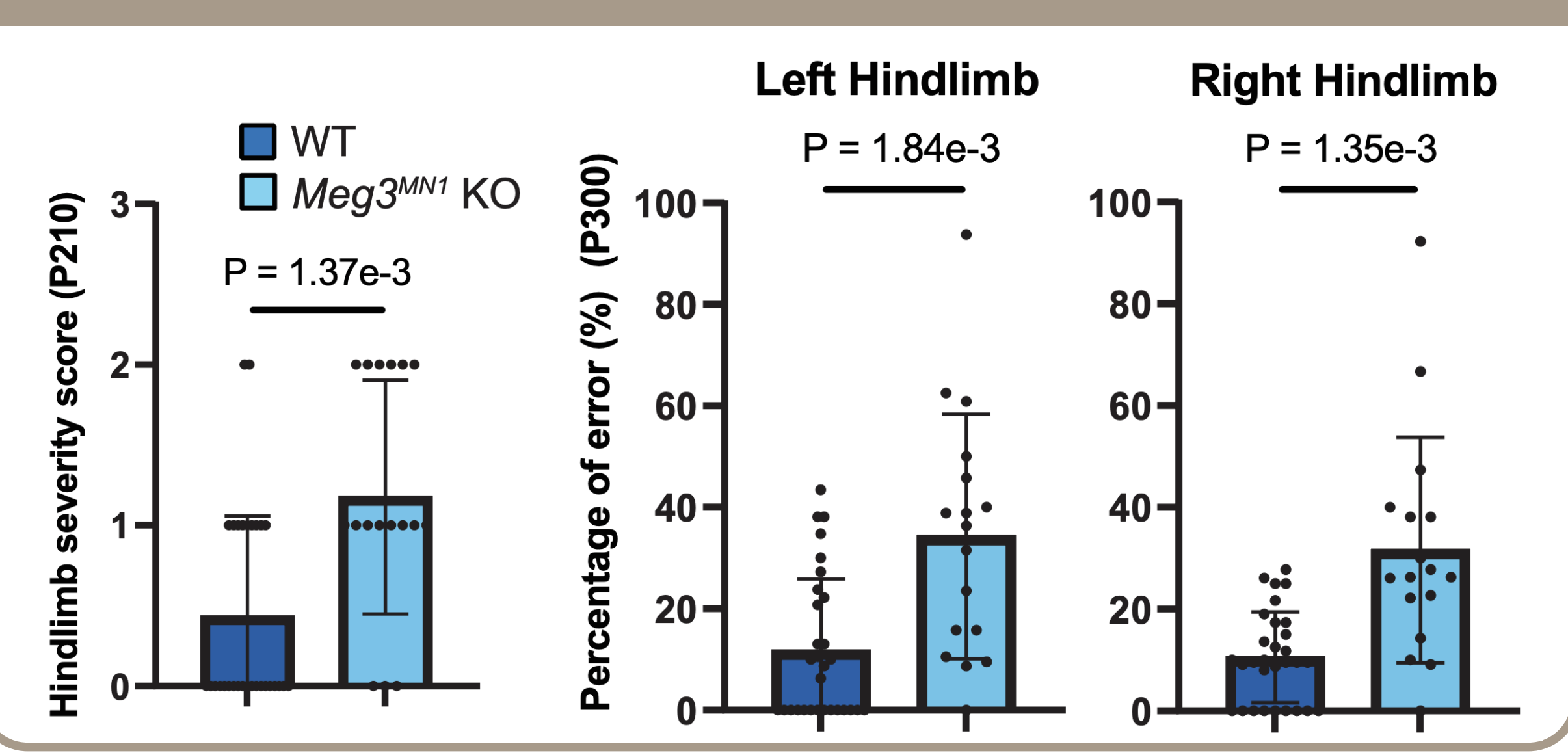
Loss of Meg3^{MN1} reduces ELAVL chromatin association



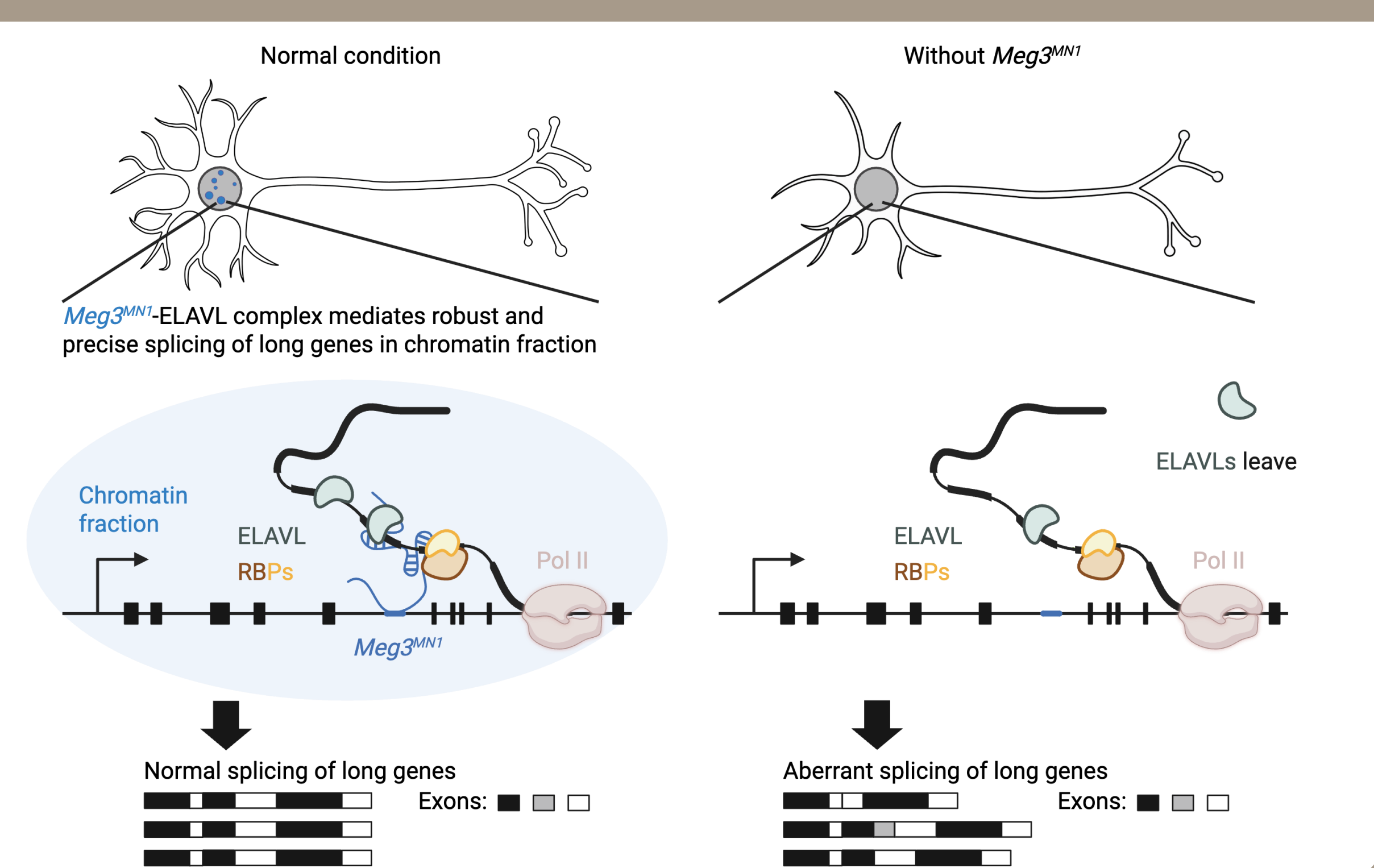
Preferentially disrupts splicing of Meg3^{MN1}-target long gene



Meg3^{MN1} loss is associated with motor impaired



Working Model



Questions

Can a neuronal lncRNA coordinate RBP's to support the processing of long neuronal genes?

Conclusions

1. Meg3^{MN1} targets long neuronal loci and associates with neuronal ELAVL proteins.
2. Meg3^{MN1} organizes ELAVL-enriched nuclear assemblies and supports ELAVL chromatin association and long-gene RNA processing.
3. Loss of Meg3^{MN1} causes long-gene splicing defects and neuronal dysfunction, highlighting a lncRNA-based mechanism for maintaining neuronal homeostasis.

Acknowledgements

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